

CALRIG · DESIGN PRECIS

CalRig design precis

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/calrig		
Source	docs/02-concept.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.7	2026-10-01	Amish Chadha	Budget treated as a value-engineering target; cost wording only, no number changed
0.6	2026-10-01	Amish Chadha	Constructable design (CLR-DDR-003): front frame, four latches, back-wall glands and bulkheads, drain, 9 mm base; numbers from CLR-CAL-001 v0.4
0.5	2026-09-26	Amish Chadha	Budget approved by Amish; \$412 budget covers the priced BOM
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). Larger inner Peltier sink, budget \$400, mass limit 14 kg; numbers from CLR-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	TRL 2 review items adopted for TRL 3 work (CLR-DDR-001); numbers replaced by CLR-CAL-001 results; full jacket, 120 mm fan, sleeved bubbler with trace-heated line, 600 x 500 mm base
0.3	2026-09-25	Amish Chadha	

Populate to TRL 2 (how it works, components, first-order numbers, safety, open questions)

0.1	2026-09-25	Amish Chadha	Initial scaffold
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CalRig design precis

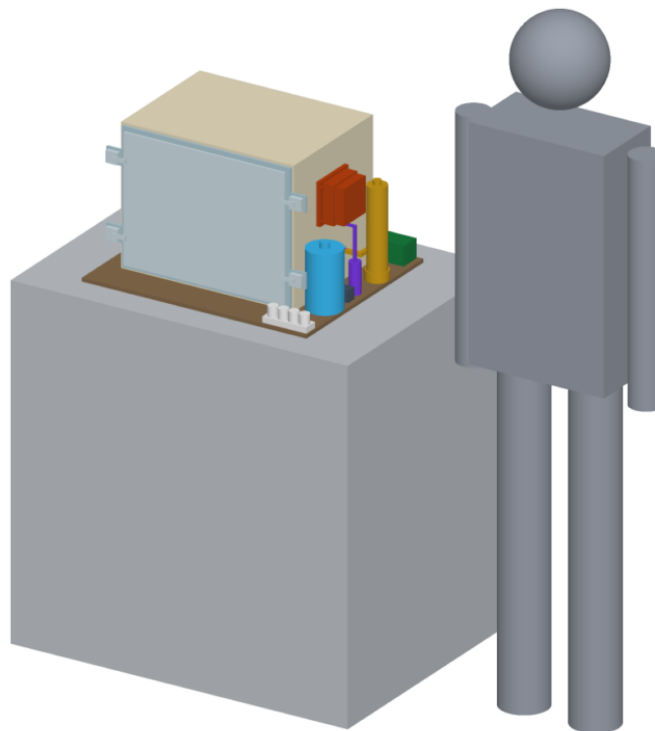
■ Summary

CalRig is a 36 L insulated acrylic chamber on a bench-top base. A Peltier heat pump sets the temperature, a heated bubbler and a silica gel dryer set the humidity, a HEPA loop gives clean air and a controlled smoke decay for particle tests, and a reference cluster (two Sensirion SHT45 temperature and humidity sensors, a Sensirion SCD30 CO₂ sensor and a collocated Sensirion SPS30 particle sensor) sits among six sensors under test. A small controller steps through set points and logs everything; a laptop script fits a correction for each sensor and writes a dated record. The calculations in CLR-CAL-001 v0.4 show that it meets twelve of the eighteen requirements in CLR-REQ-001. Making the design buildable (CLR-DDR-003, open for Amish's review) added a front frame, bulkhead fittings, cable glands, a drain and printed holders, so the estimated cost of the constructable design is about \$439 against the \$412 value-engineering target (\$27 over; see the value engineering section of CLR-DEC-001). How to build the prototype is in the build plan, CLR-BLD-001. It still depends on a collocation site for its particle reference.

CalRig

github.com/BoujeeEnjinia1701/calrig

CONCEPT, NOT FOR FABRICATION



Seen from the front right and above, 24 deg elevation. Grey: Lab bench, Person, 1.75 m for scale

Figure 1. CalRig massing model on a lab bench, with a 1.75 m person for scale. Concept, not for fabrication.

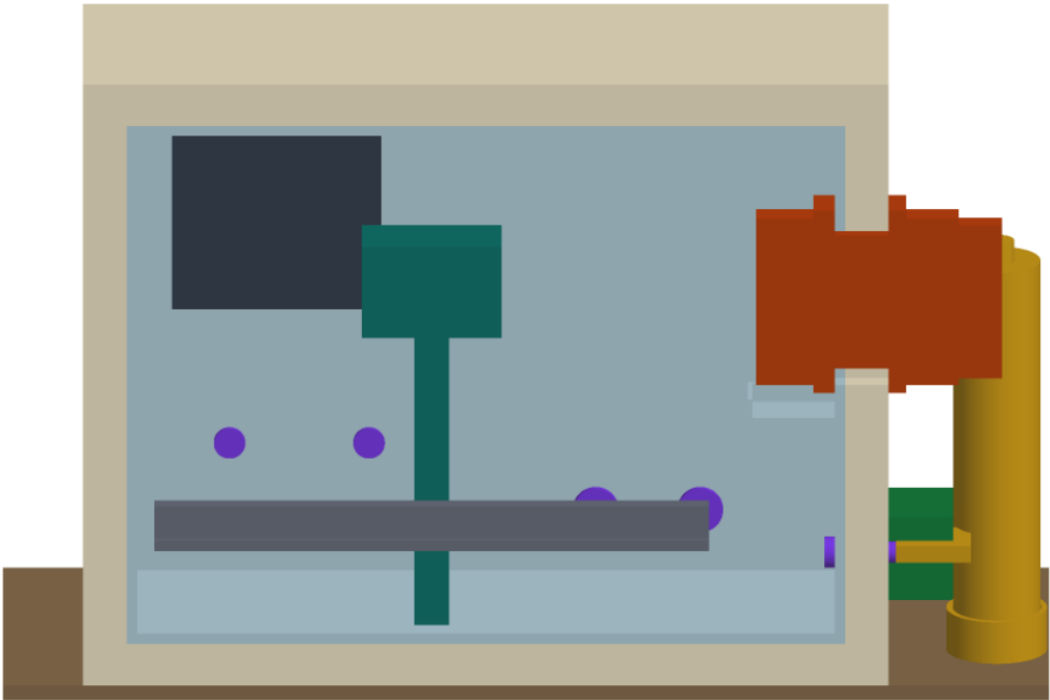
■ How it works

1. **Load.** Up to six sensor heads sit in the bays of a tray inside the chamber. Their leads pass through two sealed glands in the back wall to a USB and I2C hub on the controller. The door is closed and latched.
2. **Check the references.** Before a campaign (monthly is proposed), the two SHT45 references are checked at four saturated salt fixed points (11, 33, 75 and 84 % RH at 25 °C; [Greenspan, 1977](#)) and in an ice bath at 0 °C. The SPS30 transfer sensor is collocated at a regulatory or research monitor for 30 days or more, and replaced in the chamber only if it meets the US EPA PM2.5 targets there.
3. **Temperature and humidity sweep.** The controller steps through set points, by default 20 °C and 40 °C at 40 % and 85 % RH, matching the EPA enhanced test conditions ([EPA FAQ](#)). The sweep runs from cold to hot, because cooling is the slow direction. The Peltier assembly heats or cools. Two small pumps draw air from the chamber and return it through the bubbler (wet) and the dryer (dry) at a ratio set by PWM, with about 3 L/min in total. The internal 120 mm fan mixes the chamber. Each point settles in about 22 to 31 min, then allows 15 min for the heads to equilibrate, and is held for 30 min (CLR-CAL-001, section I).
4. **Particle run.** With the HEPA loop at full speed (about 20 L/min) the chamber is cleaned to below 2 µg/m³ in about 5 min. A syringe draws a puff of smoke from an incense stick smoldering in a cup outside the chamber and injects it through the aerosol port. The HEPA loop then runs slowly (5 L/min) and the concentration decays from about 300 to 5 µg/m³ in about 28 min, giving a continuous comparison curve. The run can be repeated at a second humidity to fit the humidity term.
5. **CO2 check (optional).** Soda lime in the dryer path gives a near-zero point. One liter of exhaled breath from a bag raises the chamber by about 1,100 ppm for a span point.
6. **Fit and record.** The laptop script compares each sensor with the references, fits a slope, offset and humidity term, computes the EPA metrics (R^2 , slope, intercept, RMSE, sensor-to-sensor precision) and writes a CSV and a one-page report per sensor serial number.

CalRig: cutaway

github.com/BoujeeEnjinia1701/calrig

CONCEPT, NOT FOR FABRICATION



Front half removed; seen from the front and above, 18 deg elevation

Figure 2. Cutaway looking from the door side: sensor tray with six example sensors, reference cluster on its mast, mixing fan and the Peltier assembly through the right-hand wall.

■ Main components

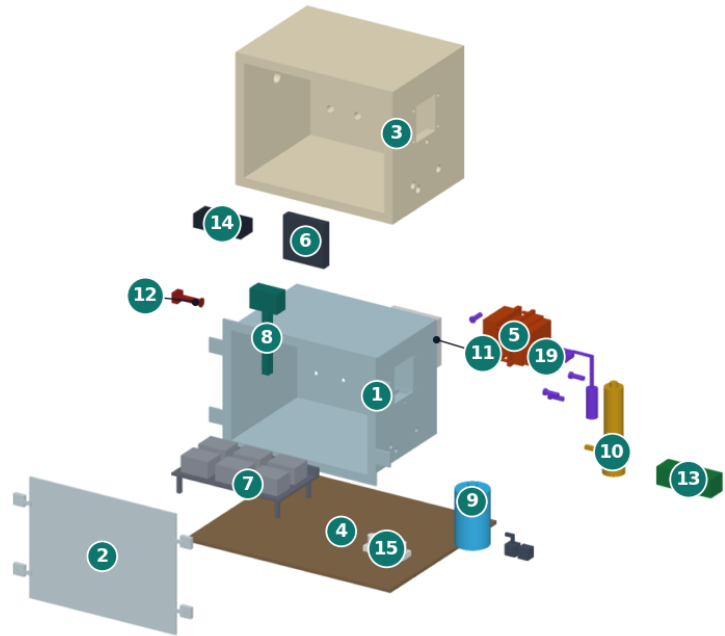
Table 1. Main components, numbered to match the BOM and the exploded view (Figure 3).

No.	Component	Role	Key choice
1	Chamber shell	36 L sealed volume (400 x 300 x 300 mm inside)	6 mm cast acrylic, solvent-welded; small enough to condition fast, large enough for six heads
2	Front door	Access and viewing	430 x 330 mm clear acrylic door on a front frame welded to the shell, silicone gasket, four draw latches
3	Insulation jacket	Cuts the chamber conductance to about 0.82 W/K	25 mm XPS on every face except the door, plus a removable door panel for hot, humid and cold points
4	Base plate	Carries the chamber and conditioners	9 mm sealed birch plywood, 600 x 500 mm, six rubber feet
5	Peltier heat pump	Heats and cools the chamber	60 W air-to-air thermoelectric assembly; the stock inner sink is replaced by a larger fin block (about 45 x 120 x 110 mm, about 0.20 K/W with its fan) so it stays above the dew point at 20 °C and 85 % RH (CLR-DDR-002)
6	Internal mixing fan	Uniform air across the bays	120 mm, 12 V, speed-controlled (80 mm was too small for R4)
7	Sensor tray	Holds six sensor heads, cable rail	Perforated sheet so air moves around the heads
8	Reference cluster	The known values	2 x SHT45 (± 0.1 °C, ± 1.0 % RH typical, Sensirion), SCD30 ($\pm(30 \text{ ppm} + 3 \text{ %})$, Sensirion), SPS30 collocated transfer sensor (± 10 % precision, Sensirion)
9	Humidifier bubbler	Wet air	0.3 L of distilled water in a foam-sleeved jar with a 10 W heater pad, held about 3 K above chamber air; trace-heated outlet line so the wet air does not condense on the way; no ultrasonic mist, which would add particles
10	Dryer column	Dry air	Indicating silica gel, about 500 g, regenerated in an oven
11	HEPA scrubber loop	Zero air and controlled smoke decay	H13 class filter cartridge with a variable-speed fan, joined to the chamber by two bulkheads in the back wall
12	Aerosol injection port	Smoke entry	Luer port with a ball valve; the smoke source stays outside
13	Controller and power board	Control, logging and protection	ESP32 class board, Peltier H-bridge, pump and fan MOSFETs, microSD, independent thermal cut-off
14	Power supply	12 V, 10 A	Certified external brick; no mains wiring in the rig
15	Salt fixed-point jars	Humidity reference checks	LiCl, MgCl ₂ , NaCl and KCl slurries in sealed jars

CalRig: exploded viewgithub.com/BoujeeEnjinia1701/calrig

CONCEPT, NOT FOR FABRICATION

- 1 Chamber shell, 6 mm acrylic, 36 L
- 2 Front door with gasket and latches
- 3 Insulation jacket (XPS, all faces but the door)
- 4 Base plate, 600 x 500 mm
- 5 Peltier heat pump, 60 W
- 6 Internal mixing fan, 120 mm
- 7 Sensor tray, six bays
- 8 Reference cluster (SHT45 x2, SCD30, SPS30)
- 9 Humidifier bubbler, sleeved, heated line
- 10 Dryer column (silica gel)
- 11 HEPA scrubber loop
- 12 Aerosol injection port and valve
- 13 Controller and power board
- 14 12 V power supply (external)
- 15 Salt fixed-point jars
- 19 Bulkhead fittings, glands and drain



Seen from the front right and above, 24 deg elevation; numbers match bom/bom.csv

Figure 3. Exploded view. Callout numbers match Table 1 and bom/bom.csv .

■ Key numbers

All values come from CLR-CAL-001 v0.2, which gives the assumptions and the script tags. They are calculations, not measurements.

Table 2. Chamber and process figures.

Quantity	Value	Basis
Internal volume	36.0 L	400 x 300 x 300 mm
Chamber conductance	0.82 W/K with door panel; 1.17 W/K without	Full jacket (U 1.01 W/(m ² ·K)), 20 % for thermal bridges, conditioning loop 0.06 W/K
Internal heat gains	6.7 W	Fans 3.9 W, six heads 2.4 W, references 0.4 W
Water to go from 20 to 85 % RH at 40 °C	1.8 g	1.19 g in the air, about 0.59 g into the acrylic surface
Settling per set point	22 to 31 min, plus 15 min for the heads	Loop time constant 12 min; full-drive thermal ramps
Four-point sweep plus particle run	about 5.5 h	Cold to hot, 30 min holds
Decay 300 to 5 µg/m ³	about 28 min	HEPA 5 L/min, deposition 0.2 per hour
CO ₂ span from 1 L of breath	about +1,100 ppm	Exhaled air about 4 % CO ₂
Peak electrical power	about 90 W, 7.5 A at 12 V	Peltier 59 W, heaters 13 W, pumps, fans, controller, sensors
Overall size and mass	600 x 500 x 371 mm; about 13.8 kg	Parametric model <code>cad/src/model.py</code>

Table 3. Temperature range with one 12706-class module at 12 V (CLR-CAL-001, section B).

Room	Lowest chamber temperature	Heating to 40 °C
15 °C	-2.0 °C	13.9 W needed, about 80 W available
22 °C	4.1 °C	8.1 W needed, about 86 W available
25 °C	6.7 °C	Met
30 °C	11.1 °C	Met

The revised R1 (10 °C in rooms of 15 to 25 °C) is met, with a 3.3 K margin at 25 °C. Cooling to 10 °C is slow, about 2.5 h in a 25 °C room, so cold points are a separate run.

Condensation at the hot, humid point. At 40 °C and 85 % RH the dew point is 37.0 °C. The inner face of the clear door in a 22 °C room runs at about 33 °C and condenses; with the door panel fitted it runs at about 38.2 °C, a 1.2 K margin (0.5 K in a 15 °C room). The door panel is therefore required for that point, and the wet air line is trace-heated.

Condensation on the Peltier sink. Holding 20 °C in the 22 °C reference room still needs about 8 W of cooling, mostly to remove the internal gains. With the stock inner sink (0.45 K/W) the sink ran about 1.2 K below the 17.4 °C dew point of air at 85 % RH and would condense about 8 g/h against about 0.7 g/h from the bubbler. The larger inner fin block (about 0.20 K/W, CLR-DDR-002) runs at about 18.3 °C, 0.9 K above the dew point, so the point holds in rooms up to about 28 °C (R2 met). A drip tray and drain stay in the design for cold points.

Particle reference. The chamber cannot create a traceable mass reference. It carries one from the field: a transfer SPS30 that has been collocated with a regulatory or research monitor. Chamber results then describe how each sensor compares with that transfer sensor in combustion smoke at known humidity. Reports must state the aerosol type, because optical sensors respond differently to dust, salt and smoke.

■ Key design choices

These were decided by Amish on 2026-09-25 (go with recommendation; CLR-DDR-001 and CLR-DDR-002).

- **Transfer references instead of certified instruments.** Certified reference instruments cost thousands of dollars. CalRig uses good digital sensors, checked against physical fixed points (salts, ice) and field collocation. This keeps the rig near its value-engineering target and makes the uncertainty chain explicit.
- **Heated bubbler, not an ultrasonic mister.** Ultrasonic misters make mineral particles that would corrupt particle tests.
- **Smoke decay, not a nebulizer.** A decay curve covers the whole range in one run with no dilution hardware. Every report states the aerosol type (CLR-DDR-001 A4).
- **Accept the cooling limit.** One Peltier module; 10 °C only in rooms at 25 °C or below, and the lowest point stated in each record for warmer rooms (CLR-DDR-001 A3).
- **Insulated door panel** for hot, humid and cold points (CLR-DDR-001 A6).
- **Larger inner Peltier sink** (about 0.20 K/W) so the 20 °C, 85 % RH point holds in normal rooms (CLR-DDR-002).
- **Mass limit of 14 kg.** CalRig is a bench rig, so the 12 kg limit was relaxed rather than thinning the acrylic (CLR-DDR-002).
- **Certified temperature probe if the cost target allows.** It does not yet (the BOM is already above the \$412 value-engineering target), so R5 stays at risk (CLR-DDR-002).
- **External 12 V supply.** Keeps mains voltage out of a box that holds water.
- **Temperature and humidity first, CO2 optional, no toxic gases.** NO2 sensors, including AirStreet's, are calibrated by field collocation only (CLR-DDR-001 A5).

■ Links to other projects

- **AirStreet** describes itself as "calibrated on CalRig" for PM2.5 and NO2. CalRig covers PM2.5 and the temperature and humidity terms; AirStreet's README already states that NO2 is calibrated by field collocation.
- **HeatMap Node** (heat and humidity), **DustBadge** (dust) and **FieldNode** sensor heads are intended users. DustBadge measures respirable dust, so a smoke-based chamber result will not transfer directly to mineral dust.

■ Safety

Safety: The Peltier hot-side heat sink can reach 60 to 70 °C; keep it guarded and away from the acrylic. An independent thermal cut-off opens the heater circuit at 50 °C chamber air or 70 °C hot side. The 12 V bus carries up to about 7.5 A: fuse it at 10 A at the supply and use rated wire. The bubbler heater pad and the line trace each have a thermistor and are switched off by the same cut-off. Use only a certified external supply; never wire mains into the rig, which contains water.

Safety: Test smoke contains fine particles and combustion gases, including carbon monoxide in small amounts. Light the incense in a cup outside the chamber, in a ventilated room, and run the HEPA loop until the reference reads below 5 µg/m³ before opening the door. No toxic calibration gases are used in this version.

Safety: Soda lime is corrosive and lithium chloride is harmful if swallowed. Wear gloves and eye protection when filling cartridges and salt jars, label them, and keep them away from children. Cast acrylic is combustible: keep heaters and hot surfaces off the shell. Keep electronics below and outside the chamber so condensation cannot reach them. Fans have guards. Exhaled air for CO₂ checks goes through a single-user bag and filter.

■ Open questions

- ☒ Value-engineering target: \$412 in parts, a hypothetical control target set by Amish on 2026-09-26 (CLR-DDR-002). Estimated cost of the constructable design: \$439 (\$27 over the target); savings worth trying are in the value engineering section of CLR-DEC-001.
- ☐ Particle reference: which collocation site (regulatory monitor, university, AQ-SPEC style program) will host the transfer SPS30, and how often it returns there (awaiting Amish).
- ☐ Inner fin block: confirm a part that reaches about 0.20 K/W with the inner fan.
- ☐ Report format: what a city or funder would accept as evidence.